Pratyush Saxena

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EDUCATION

Cornell University - Bachelor's in Computer Science, Minor in Artificial Intelligence **Ithaca, NY | Aug 2025 - May 2028**
GPA: 3.9/4.0 | Relevant Coursework: Data Structures, Linear Algebra, Discrete Math, Object-Oriented and Functional Programming

Thomas Jefferson High School for Science and Technology - High School Diploma **Alexandria, VA | Aug 2021 - Jun 2025**
GPA: 4.0/4.0 | ACT: 35/36 | PSAT: 1500/1520 | Relevant Coursework: Artificial Intelligence, Computer Vision, App Development

EXPERIENCE

Splunk (Cisco) - Software Security & Automation Engineering Intern **San Jose, CA | May 2026 - Aug 2026**
- Built an MCP tool to normalize freeze/scheduling data so agents can determine maintenance time blocks for 15,000+ stacks.
- Reduced query latency by 100ms for Splunk's source of truth by replacing an HTTP cache with direct Redis access in Go.
- Integrated cybersecurity and code style guide standards across 15+ repositories to improve AI-generated code quality.

Alpheva AI - Product Analytics & Engineering Intern **New York City, NY | Nov 2025 - Jan 2026**
- Built a React Native Reports dashboard with cash-flow visualizations and category-level breakdowns (e.g., dining, travel, subscriptions), improving users' insight into spending behavior and contributing to ~$6,500 in average annual user savings.
- Helped design investor pitch decks used to secure partnerships with institutions including OpenAI, Google, and Amazon.
- Drove go-to-market execution by analyzing 6 competing fintech products (features, release progress, and social media marketing), producing live product demo videos, and conducting targeted outreach to ~1,000 venture capital investors.

Biostate AI - AI Research Intern **Houston, TX | Aug 2025 - Oct 2025**
- Worked in a human-in-the-loop ML research workflow by reviewing AI-generated analyses, validating statistical results, manually correcting figures, and editing LLM-generated manuscript drafts to meet scientific and publication standards.
- Applied CNN-based analysis and LLM tools to immunology datasets (CD8$^+$ PD-L1$^+$ cells in murine models) and co-authored 5+ AI-assisted manuscripts submitted to peer-reviewed journals, including *Genome Biology*.

NASA - Software Engineer Intern **Greenbelt, MD | Jun 2024 - Aug 2024**
- Developed Python software for laser-based wireless power transfer between in-orbit satellites as a proof-of-concept mission.
- Led a 5-person team on two projects: one focused on coding an OpenCV-based computer vision algorithm for laser-point tracking, and the other on developing Inertial Measurement Unit (IMU)-based software for in-orbit satellite stabilization.
- Presented technical performance metrics, including ~98% laser-aiming accuracy, to 100+ NASA engineers and scientists.

PROJECTS

Baya: AI-Powered Interior Design Platform **May 2026 - Jul 2026**
- Built an AI interior design platform combining Claude and 4 fal.ai image models to generate photorealistic room redesigns from a single photo, chaining up to 7 sequential edits while preserving room architecture and providing live links to products.
- Integrated cross-provider fallback between Claude Sonnet and Haiku and between fal.ai and Replicate, distributed concurrency control using Upstash, and a pgvector-based product matching system seeded from eBay and Amazon APIs.

Drizzle: Autoimmune Disorder Social Networking App **Mar 2026 - Jun 2026**
- Built a full-stack cross-platform social media app for individuals with autoimmune disorders using React Native and Next.js.
- Implemented scalable features including Supabase Auth (OAuth), real-time messaging, fine-grained privacy controls, push notifications, infinite-scrolling feeds, and Cloudflare R2 image uploads with cloud-backed synchronization.

4Sight: AI-Driven Insider Trading Monitor and Interpreter **Nov 2025 - Feb 2026**
- Built a Python-based web scraper that collects and parses SEC Form 4 insider trading filings, tested across 50+ tickers.
- Developed an NLP model that analyzes insider activity alongside global news to generate potential explanations for trades.

T-REX: Energy-Harvesting Piezoelectric Floor Tile **Jul 2025 - Sep 2025**
- Prototyped a floor tile that harvests electricity from mechanical and sonic energy using piezoelectric crystals, achieving roughly 1.4 mW of power per step per dollar, over 30x more cost-efficient than competing drum-harvester technology.
- Designed an AI resonance-tuning system that predicts ambient vibration and sound frequencies and adjusts pressure on the piezoelectric crystal via a motorized brace to keep the tile near peak resonance for maximum energy output.

SkIntel: CNN-Based Skin Cancer Screening Computer Vision Model **Feb 2025 - Jun 2025**
- Coded a convolutional neural network in Python with an MIT PhD-led team to detect skin cancer from lesion images.
- Achieved high accuracy (AUC score of 0.93) using smartphone-quality images and presented results to an audience of 150+.

SKILLS & HONORS

Technical: Python, Java, Go, SQL, JavaScript, Flask, React, NumPy, OpenCV, TensorFlow, phpMyAdmin, Excel, Onshape, ArcGIS

Extracurricular Activities: Generative AI Club, Physical Intelligence Club, Association for Computer Science Undergraduates

Academic: National Merit Scholar Finalist, National Honor Society, Science Olympiad States, MathCounts Chapter Invitationals

Extracurricular: Eagle Scout, TSA Software Development Nationals, Congressional Debate Metrofinals, Tae Kwon Do Black Belt